

Question	Answer	Mark	Mark scheme	Additional guidance
21	-6, 3.5	P1	for process to find gradient of PR or QR, eg $\frac{k+3-5}{k-4} \left(= \frac{k-2}{k+4} \right)$ or $\frac{k+3-6}{k-6} \left(= \frac{k+9}{k-6} \right)$ or $\frac{5-(k+3)}{-4-k} \left(= \frac{2-k}{-4-k} \right)$ or $\frac{-6-(k+3)}{6-k} \left(= \frac{-9-k}{6-k} \right)$ OR for start of process to use Pythagoras, eg $PR^2 = (k-4)^2 + (k+3-5)^2$ or $RP^2 = (-4-k)^2 + (-5-(k+3))^2$ or $QR^2 = (k-6)^2 + (k+3-6)^2$ or $RQ^2 = (6-k)^2 + (-6-(k+3))^2$	Condone missing bracket for first P1 only Stating eg Grad $PR = \frac{k-4}{k+3-5}$ Or grad $QR = \frac{k-6}{k+3-6}$ scores P0 but check SC
		P1	for forming a correct equation, eg $\frac{k+3-5}{k-4} \times \frac{k+3-6}{k-6} = -1$ or $\frac{k-2}{k+4} \times \frac{k+9}{k-6} = -1$ or $\frac{5-(k+3)}{-4-k} \times \frac{-6-(k+3)}{6-k} = -1$ or $\frac{2-k}{-4-k} \times \frac{-9-k}{6-k} = -1$ oe OR $(k-4)^2 + (k+3-5)^2 + (k-6)^2 + (k+3-6)^2 = (-6-5)^2 + (6-4)^2$ or $(k+4)^2 + (k-2)^2 + (k-6)^2 + (k+9)^2 = (-11)^2 + 10^2$	
		P1	for writing in the form $ak^2 + bk + c (= 0)$, eg $2k^2 + 5k - 42 (= 0)$ or $4k^2 + 10k - 84 (= 0)$	
		P1	(dep P3) for factorising, eg $(k+6)(2k-7) (= 0)$ or $(2k+12)(2k-7)$ or $(k+6)(4k-14)$ OR use of formula, eg $\frac{-5 \pm \sqrt{5^2 - 4 \times 2 \times -42}}{2 \times 2}$ or $\frac{-10 \pm \sqrt{10^2 - 4 \times 4 \times -84}}{2 \times 4}$	
		A1	cao SCB4 for correct answer coming from consistent use of reciprocals of gradients	